```
#include <stdio.h>
#include <conio.h>
#include <graphics.h>

void ellipse_midpoint(int xc, int yc, int rx, int ry)
{
   int x = 0;
   int y = ry;

   long rx2, ry2;
   long dx, dy;
   double p1, p2;

   rx2 = (long)rx * rx;
   ry2 = (long)ry * ry;

   dx = 2 * ry2 * x;
   dy = 2 * rx2 * y;

   /* Region 1 */

   p1 = ry2 - (rx2 * ry) + (0.25 * rx2);

   while (dx < dy)
   {
```

```
    putpixel(xc + x, yc + y, WHITE);
    putpixel(xc - x, yc + y, WHITE);
    putpixel(xc + x, yc - y, WHITE);
    putpixel(xc - x, yc - y, WHITE);

    if (p1 < 0)
    {
        x++;

        dx = dx + 2 * ry2;

        p1 = p1 + dx + ry2;
    }
    else
    {
        x++;
        y--;

        dx = dx + 2 * ry2;
        dy = dy - 2 * rx2;

        p1 = p1 + dx - dy + ry2;
    }
}
```

```
/* Region 2 */

p2 = ry2 * (x + 0.5) * (x + 0.5)
   + rx2 * (y - 1) * (y - 1)
   - (double)rx2 * ry2;

while (y >= 0)
{
    putpixel(xc + x, yc + y, WHITE);
    putpixel(xc - x, yc + y, WHITE);
    putpixel(xc + x, yc - y, WHITE);
    putpixel(xc - x, yc - y, WHITE);

    if (p2 > 0)
    {
        y--;

        dy = dy - 2 * rx2;

        p2 = p2 + rx2 - dy;
    }
    else
    {
        x++;
        y--;
```

```
            dx = dx + 2 * ry2;
            dy = dy - 2 * rx2;

            p2 = p2 + dx - dy + rx2;
        }
    }
}

void main()
{
    int gd = DETECT, gm;

    initgraph(&gd, &gm, "C:/TURBO C3/BGI");

    ellipse_midpoint(300, 200, 150, 80);

    getch();
    closegraph();
}
```